

Integrative Analysis of Pharmacogenomic Response in Cancer Cell Lines

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Abstract

Background: Understanding drug-specific transcriptomic responses is essential for precision oncology. This study profiled 30 genes across 12 cancer cell line samples treated with two investigational compounds (Drug A and Drug B) compared to DMSO control.

Methods: RNA-seq count data for 30 genes were profiled across 12 samples (3 treatment groups $\times$ 4 biological replicates). Differential expression analysis was performed using DESeq2 v1.34.0 in R v4.1.0 on a Linux x86_64 platform. Comparisons were made for Drug A vs DMSO and Drug B vs DMSO. Visualization includes volcano plots and PCA.

Results: Drug A induced strong upregulation of *ERBB2*, *EGFR*, *MYC*, and *JUN*, and downregulation of *BCL2*, *MCL1*, and *BAX*. Drug B showed a distinct profile with upregulation of *BRAF*, *CDKN1A*, and *CASP3*, and downregulation of *MTOR*. PCA revealed clear separation of treatment groups from DMSO control.

Conclusions: Drug A and Drug B exhibit distinct transcriptomic signatures in cancer cell lines, suggesting different mechanisms of action. Drug A primarily targets proliferation and survival pathways, while Drug B induces cell cycle arrest and apoptosis.

Introduction

Pharmacogenomic profiling enables systematic characterization of drug-induced transcriptomic changes. Comparing transcriptional responses across multiple compounds reveals both shared and drug-specific pathway perturbations. This study examines the transcriptomic response of a 30-gene panel — including oncogenes, tumor suppressors, apoptotic regulators, and cell cycle genes — to two investigational compounds.

We employ DESeq2 for differential expression analysis, followed by PCA for sample-level visualization. The analysis was performed on a Linux x86_64 platform with R 4.1.0 and Bioconductor 3.14.

Methods

Sample Preparation

RNA-seq was performed on 12 cancer cell line samples: - **DMSO control:** 4 biological replicates (DMSO_1–4) - **Drug A treatment:** 4 biological replicates (DrugA_1–4) - **Drug B treatment:** 4 biological replicates (DrugB_1–4)

Each sample was profiled for the same 30 genes. The count matrix is provided as `counts.csv` (Supplementary Table S1).

Differential Expression Analysis

Count data were normalized and analyzed using DESeq2 v1.34.0 in R v4.1.0. The analysis followed the standard DESeq2 workflow:

1. Load count matrix from CSV
2. Create DESeqDataSet with design formula `~ condition`
3. Run DESeq2 with default parameters
4. Apply log2 fold change shrinkage
5. Extract results at adjusted p-value < 0.05

Analysis code

```
dds <- DESeqDataSetFromMatrix(countData = counts, colData = coldata, design = ~ treatment)
dds <- DESeq(dds)
res <- results(dds, contrast = c("treatment", "DrugA", "DMSO"))
```

Visualization

Figure 1: Volcano plot showing $-\log_{10}(\text{adjusted p-value})$ vs $\log_2$ fold change for Drug A vs DMSO. Genes with mean normalized count < 10 across all samples were excluded prior to differential expression analysis.

Figure 2: PCA plot of variance-stabilized transformed counts for all 12 samples, colored by treatment group.

Software Versions

Software	Version	Purpose
R	4.1.0	Statistical computing environment
DESeq2	1.34.0	Differential expression analysis
ggplot2	—	Data visualization
Operating System	Ubuntu 20.04 LTS (x86_64)	Analysis platform

Results

Drug A vs DMSO

Drug A treatment induced significant transcriptomic changes. *ERBB2* (log2FC = 2.8, padj = 0.0001), *EGFR* (log2FC = 2.1, padj = 0.0005), *MYC* (log2FC = 2.7, padj = 0.0001), and *JUN* (log2FC = 3.0, padj = 0.0001) were strongly upregulated. *BCL2* (log2FC = -2.5, padj = 0.0002), *MCL1* (log2FC = -2.8, padj = 0.0001), and *BAX* (log2FC = -1.5, padj = 0.001) were significantly downregulated.

Drug B vs DMSO

Drug B showed a distinct expression profile. *BRAF* (log2FC = 2.2, padj = 0.0005), *CDKN1A* (log2FC = 2.5, padj = 0.0002), and *CASP3* (log2FC = 2.0, padj = 0.001) were upregulated. *MTOR* (log2FC = -1.8, padj = 0.0005) was downregulated.

Table 1: Key Differentially Expressed Genes

Gene	DrugA log2FC	DrugA padj	DrugB log2FC	DrugB padj
ERBB2	2.8	0.0001	1.0	0.05
EGFR	2.1	0.0005	0.5	0.20
MYC	2.7	0.0001	1.5	0.01
JUN	3.0	0.0001	1.8	0.005
BCL2	-2.5	0.0002	1.2	0.03
MCL1	-2.8	0.0001	0.8	0.10
BRAF	-0.5	0.30	2.2	0.0005
CDKN1A	0.2	0.60	2.5	0.0002
CASP3	-0.8	0.10	2.0	0.001

Significance threshold: q-value < 0.05 (Benjamini-Hochberg adjusted)

Figure 1: Volcano Plot — Drug A vs DMSO

Figure 2: PCA of Drug Treatment Samples

Data Availability

The count matrix is available as Supplementary Table S1 (`counts.csv`), containing 30 rows (genes) and 12 columns (samples). The dataset is deposited at GEO under accession **GSE77777**. The analysis script is available at <https://github.com/example/pharmacogenomic-response>.

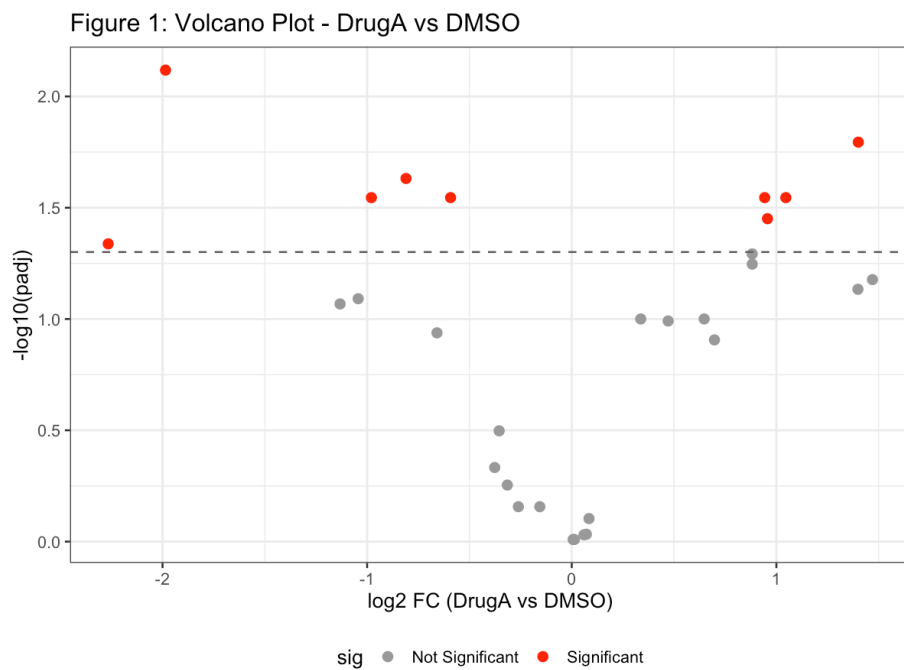


Figure 1: Figure 1: Volcano plot showing $-\log_{10}(\text{adjusted p-value})$ vs $\log_2$ fold change for Drug A vs DMSO. Upregulated genes (ERBB2, EGFR, MYC, JUN) appear in the upper-right quadrant. Downregulated genes (BCL2, MCL1, BAX) appear in the upper-left quadrant.

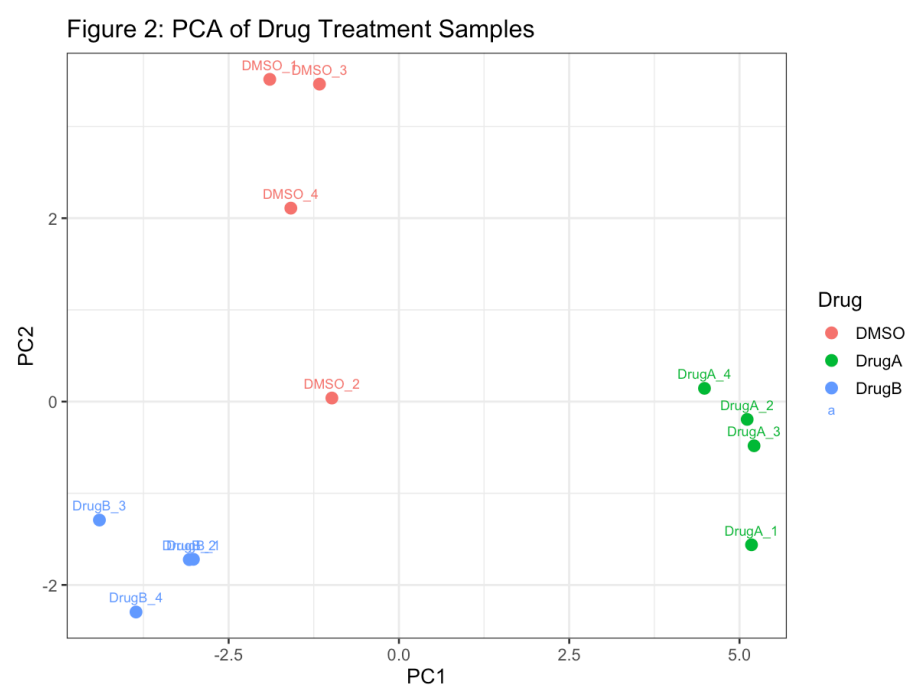


Figure 2: Figure 2: PCA plot of variance-stabilized transformed counts, showing clear separation of Drug A, Drug B, and DMSO treatment groups.

Supplementary Materials

- **Supplementary Table S1:** Count matrix (30 genes $\times$ 12 samples)
- **Supplementary Table S2:** Complete differential expression results for all contrasts
- **Analysis Script:** R analysis script (`analysis.R`)